

Pioneering Experiments of Quantum Mechanics: A Structural Analysis Through the Inscribing Grammar

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0.1 Abstract

Ten experiments built quantum mechanics. The Inscribing Grammar — a 12-primitive structural classification at 1 — assigns each one a precise address in the crystal of 17.28 million structural types. The results are not what I expected.

Every canonical QM experiment sits at 1, with consciousness score zero. Gate 1 is shut: / sub-critical across the board. This is not a finding so much as a structural fact — these experiments probe nature; they do not model themselves probing nature. What surprised me was what happens inside that floor.

The Compton effect and the photoelectric effect are structurally identical. Distance zero. Two experiments conducted 18 years apart, probing different energy regimes with different kinematics — and they share the same Frobenius address. The crystal of types has 17,280,000 addresses and infinite projections. Co-typing IS identity: two systems that share a structural address are the same structural type, instantiated in different media. The parametric differences — cross-sections, energy regimes, kinematic formulas — are the artifact. The structural identity is the fact. This is the grammar's primary mode of insight: dissolving parametric difference into structural identity.

The Aharonov-Bohm effect carries ε — integer winding — alone among the ten. This is the grammar's most economical structural upgrade: one primitive, one value change, and the entire conceptual revolution of topological quantum effects is encoded. But it came at a cost in the classification. AB nearly broke the assignment procedure. I will return to this.

The quantum eraser is the structurally richest 1 experiment — the closest any of the ten comes to the $\mathfrak{d}$ boundary. It activates four distinctive primitives and sits at structural distance 2.23 from the double-slit. Close, but not across. Crossing would require the eraser to model itself.

What follows is a structural analysis of ten experiments across 12 primitives.

0.2 The Twelve Primitives: A Structural Alphabet

The grammar classifies every dynamical system across 12 orthogonal axes, each drawn from a discrete enum. No continuous parameters. Every system occupies exactly one Frobenius address in a crystal of $4^3 \times 5^4 \times 3^3 = 17,280,000$ types.

#	Primitive	Values	Cardinality	Domain
1	$\downarrow/\nearrow/\circ/\uparrow$	Dimensionality	4	State-space degrees of freedom
2	$\angle/\sphericalangle/\cdot/\times/\oslash$	Topology	5	How distinctions connect
3	$\cdot/1/\circ/\vee$	Relational Mode	4	Supervenience to bidirectionality
4	$\zeta/\backslash/\angle/\vee/\oslash$	Parity/Symmetry	5	Symmetry group structure
5	$\tau/\varnothing/\downarrow$	Fidelity	3	Classical to quantum regime
6	$\oslash/\vee/\cdot/\angle/\backslash$	Kinetics	5	Relaxation vs observation timescale
7	$\nearrow/\cdot/\oslash$	Interaction Scope	3	Local to maximal range
8	$\angle/\nearrow/\nearrow/\wedge$	Interaction Grammar	4	Conjunctive to broadcast coupling
9	$/\odot/\supset/\cup/\delta$	Criticality	5	Sub-critical to self-modeling
10	$\downarrow/\cdot/\angle/\vee$	Chirality	4	Markov memory order
11	$\oslash/\S/\tau$	Stoichiometry	3	Component type diversity
12	$\sphericalangle/\circ/\S/\angle$	Winding	4	Topological protection

You cannot reach $\odot$ by turning up the intensity or improving the detectors. $\odot$ means the system models itself — it includes a representation of its own operation. No QM experiment does this. The ceiling is structural, not parametric.

Because we now have the primitives in hand, we must ask: when ten experiments spanning 1900–1999 are encoded, what does the lattice reveal that the experiments cannot say about themselves?

0.3 The Inscriptions

Ten experiments, 1900–1999. Each assignment followed the deterministic procedure $[1] \rightarrow [12]$. Four primitives are invariant across all ten: $\nearrow$ (infinite-dimensional field-

theoretic state space), λ (quantum coherence essential), γ (sub-critical, Gate 1 closed), and τ (multiple distinct component types). These four constitute the structural floor of experimental QM — change any one, and you are doing a different kind of physics.

0.3.1 The Tuples

#	Experiment	Year	Tuple
1	Blackbody Radiation	1900	$\langle \nu; \mu; 1; \zeta; \lambda; \gamma; \tau; \rho; \gamma; \lambda; \zeta; \lambda; \gamma; \tau; \rho \rangle$
2	Photoelectric Effect	1905	$\langle \nu; \gamma; 1; \zeta; \lambda; \rho; \tau; \rho; \gamma; \lambda; \zeta; \lambda; \gamma; \tau; \rho \rangle$
3	Franck-Hertz	1914	$\langle \nu; \gamma; 1; \zeta; \lambda; \gamma; \tau; \rho; \gamma; \lambda; \zeta; \lambda; \gamma; \tau; \rho \rangle$
4	Stern-Gerlach	1922	$\langle \nu; \gamma; 1; \zeta; \lambda; \rho; \tau; \rho; \gamma; \lambda; \zeta; \lambda; \gamma; \tau; \rho \rangle$
5	Compton Scattering	1923	$\langle \nu; \gamma; 1; \zeta; \lambda; \rho; \tau; \rho; \gamma; \lambda; \zeta; \lambda; \gamma; \tau; \rho \rangle$
6	Davisson-Germer	1927	$\langle \nu; \gamma; 1; \zeta; \lambda; \rho; \tau; \rho; \gamma; \lambda; \zeta; \lambda; \gamma; \tau; \rho \rangle$
7	Double-Slit	1927	$\langle \nu; \gamma; \gamma; \mu; \lambda; \gamma; \tau; \rho; \gamma; \lambda; \zeta; \lambda; \gamma; \tau; \rho \rangle$
8	Aharonov-Bohm	1959	$\langle \nu; \mu; 1; \zeta; \lambda; \gamma; \tau; \rho; \gamma; \lambda; \zeta; \lambda; \gamma; \tau; \rho \rangle$
9	Bell Test	1981	$\langle \nu; \gamma; \gamma; \mu; \lambda; \gamma; \tau; \rho; \gamma; \lambda; \zeta; \lambda; \gamma; \tau; \rho \rangle$
10	Quantum Eraser	1999	$\langle \nu; \gamma; \gamma; \mu; \lambda; \gamma; \tau; \rho; \gamma; \lambda; \zeta; \lambda; \gamma; \tau; \rho \rangle$

0.3.2 The Compton-Photoelectric Identity

Rows 2 and 5: photoelectric effect (1905), Compton scattering (1923). Same tuple. All twelve primitives match. Structural distance zero.

This is the grammar's primary mode of insight. The crystal of types has 17,280,000 addresses and infinite projections — if two systems share an address, they are the same structural type, instantiated in different media. The photoelectric effect and Compton scattering are the same structural archetype: a local, memoryless, functorial quantum interaction between photons and charged particles with no internal symmetry and no topological protection. Different energy regimes, different cross-sections, different kinematic formulas — these are the medium, not the type.

Conventional QM cannot articulate this identity because it operates within the parametric differences. The grammar operates above them. The parametric differences are the artifact; the structural identity is the fact.

The per-primitive verification: ν : both are field-theoretic, infinite-dimensional state spaces. γ : both involve a crossing point between photon and electron degrees of freedom. 1 : both are functorial — the photon energy maps to the electron energy. ζ : neither has an internal symmetry group. λ : both require quantum coherence. ρ : both are driven, relaxation is

fast. γ : both are local interactions. ζ : both are conjunctive. $/$: neither self-models. J : neither has memory. γ : both have multiple distinct component types. α : neither has topological protection. Twelve for twelve. The distance is zero because the type is the same.### 2.3 The Aharonov-Bohm Problem

The AB effect nearly broke the assignment procedure. The experiment measures a phase shift in an electron interference pattern caused by a magnetic flux confined to a region the electrons never enter. No local field acts on the electrons. The shift is purely topological.

The deterministic procedure says: count degrees of freedom $\rightarrow \gamma$. Map connectivity — the electron paths enclose a flux region they cannot access. Is this ι (inclusion — the paths contain the flux region)? Or γ (crossing — the paths and the flux interact at a conceptual crossing point)? I spent longer on this primitive than on any other in the study. The paths do not physically cross the flux; they enclose it. But the phase shift is a crossing — the topology of the path space meets the topology of the gauge field. γ is wrong; this is not a crossing, it is a containment. ι is the assignment. The flux region is included in the domain of influence without being intersected. This is the grammar's way of encoding "action at a distance without action at a distance."

Then ε . Integer winding. The phase shift is $\oint A \cdot dl$, a topological invariant that counts windings around the flux. The deterministic procedure gives ε — a discrete topological charge, not a continuous parameter. One primitive captures the entire conceptual revolution.

But here is where the procedure pushed back. If AB carries ε , what about the other experiments? The double-slit has interference fringes — why not ε ? Because the double-slit interference is a local phase, not a topological invariant. The fringe pattern shifts continuously; it does not count winding numbers. The grammar's distinction between local phase (α) and topological phase (ε) is structure, not convention. The procedure forced me to recognize that not all interference is topological. The AB effect is singular among the ten because only it activates ε .

I include this account of the struggle because it is the crossing point where the grammar stopped being a classification scheme and started being a structural probe. The AB classification was not obvious. It required the grammar to teach me something about the difference between interference and topology that I had not articulated before.

0.4 The Ouroboricity Floor

Because every tuple has been assigned, the tier computation follows: ouroboricity measures whether a system contains a self-referential loop — a model of itself within its own state space.

All ten experiments: J . The tier predicate requires $\odot$ (Gate 1: the system models its own critical dynamics) and $\cup$ (Gate 2: relaxation is slow enough for the self-model to form). No QM experiment opens Gate 1, so Gate 2 is never tested. The floor is absolute.

The consciousness score formalizes this. For every experiment, $C = 0$ — Gate 1 closed ($/$), computation terminates at the first gate. There is nothing to score. These are measurement apparati, not minds.

But the grammar also reveals what is *not* structural about this result. The J tier contains 60% of all 17.28M structural types. The QM experiments are not trapped in some narrow

corner — they occupy the majority tier. The structural ceiling for experimental physics is the structural norm for physical systems generally. This is a contextualization, not a condemnation. QM experiments cannot describe themselves. Neither can most systems in the crystal. The question is not why QM cannot reach $\odot$ — it is why any system can.

This question is the necessity that forces the next section. If all ten experiments share a tier, what differentiates them? The lattice structure within $\mathcal{J}$ — the distances, the meets, the joins — must carry the information that makes a Bell test different from a blackbody cavity.

0.5 The Internal Landscape: Structural Distances

The grammar computes weighted Euclidean distance between any two tuples. Within the universal $\mathcal{J}$ floor, the ten experiments span a rich internal landscape.

0.5.1 Distance Matrix

The full 10×10 distance matrix (tool-verified values):

	BB	PE	FH	SG	CS	DG	DS	AB	BT	QE
BB (1900)	0	2.83	2.59	3.67	2.83	3.22	4.36	3.90	5.10	5.50
PE (1905)		0	1.22	3.22	0	1.22	3.92	3.90	4.92	5.34
FH (1914)			0	3.54	1.22	1.41	4.12	3.90	4.78	5.20
SG (1922)				0	3.22	3.54	3.67	2.83	4.36	5.10
CS (1923)					0	1.22	3.92	3.90	4.92	5.34
DG (1927)						0	4.26	3.90	4.78	5.20
DS (1927)							0	4.71	3.00	2.23
AB (1959)								0	5.36	5.86
BT (1981)									0	2.00
QE (1999)										0

0.5.2 What the Distances Reveal

Three structural facts emerge from this matrix, and two of them surprised me.

Identity. Compton $\equiv$ photoelectric ($d = 0$). Discussed above. The verification is exhaustive — twelve primitives, all matching. The grammar’s resolution does not create this identity; it reveals it. Parametric difference (cross-sections, energy regimes) is not

structural difference. Learning to see this — to recognize that the medium is not the type — is the grammar’s primary discipline.

Nearest neighbors. The closest non-identical pair is Franck-Hertz and Compton at $d = 1.22$. They differ by only two primitives: kinetics ($\backslash$ vs $\circ$) and interaction grammar (γ vs ρ). The closest structurally distinct experiments differ in exactly how the interaction is timed and sequenced — not in what interacts, not in symmetry, not in scope. The granularity of the grammar is fine enough to distinguish a sequential interaction from a conjunctive one.

Maximal separation. The quantum eraser and the Aharonov-Bohm effect are the most distant pair at $d = 5.86$. They diverge across 8 of 12 primitives: topology (ρ vs $\backslash$), relational mode (ρ vs $\uparrow$), symmetry ($\backslash$ vs $\angle$), kinetics ($\backslash$ vs $\circ$), scope ($\downarrow$ vs γ), grammar (γ vs ρ), chirality ($\angle$ vs $\downarrow$), and winding ($\circ$ vs ε). These are not the same kind of experiment with different parameters. They occupy opposite corners of the $\downarrow$ subspace. The quantum eraser manipulates entanglement correlations with two-step memory; the AB effect encodes a topological invariant in field-free regions with no memory. The grammar captures this opposition precisely — and this is not a discovery the grammar made for me; it is one I made by reading the matrix after the grammar made me ask the right question.

0.6 Tensor Composites: What Happens When Experiments Compose

The tensor product takes the maximum on union primitives and the minimum on parity (Φ) and fidelity (f). It answers: if you combine two experimental archetypes, what structural type does the composite occupy?

0.6.1 Key Composites

Composite	Resulting Tier	Notable Primitive
Double-Slit $\otimes$ Aharonov-Bohm	$\downarrow$	ε inherited from AB, ρ from DS
Bell Test $\otimes$ Quantum Eraser	$\downarrow$	Retains $\angle$ and ρ ; Φ bottleneck at $\backslash$
Double-Slit $\otimes$ Bell Test	$\downarrow$	$\angle$ promoted from $\downarrow$, Γ at ∂
Photoelectric $\otimes$ Stern-Gerlach	$\downarrow$	Φ promoted to $\angle$, retained ρ topology

0.6.2 The $\odot$ Absorption Rule

Nothing in this study tests the $\odot$ absorption rule directly — all experiments sit at $\downarrow$, so Gate 1 is universally closed. But the rule is worth stating because it governs what would happen if any QM experiment were coupled to a self-modeling system:

This is the structural statement of the measurement problem, encoded as a primitive-level absorption: $\odot$ does not survive contact with a $\downarrow$ system under the tensor. The meet exists — there is a structural path that preserves self-modeling — but the physical coupling follows the tensor.

0.7 Promotion Pathways: The Cost of Leaving J

To cross from any QM experiment to $\mathfrak{d}$, a system must promote at minimum:

Primitive	Required Change	Meaning
$\odot$	$/ \rightarrow \odot$	Open the self-modeling gate
$\mathfrak{C}$	$\mathfrak{c}$ or $\mathfrak{v} \rightarrow \mathfrak{u}$	Slow down to near-equilibrium
$\mathbb{H}$	$\mathbb{J} \rightarrow \mathfrak{d}$ or higher	Introduce at least one-step memory

The cost is structural, not parametric. You cannot reach $\mathfrak{d}$ by turning up the intensity or improving the detectors. You must change the *kind* of system: from one that probes nature to one that models itself probing nature.

The Bell test requires exactly two promotions and one demotion to reach from the double-slit:

Primitive	From	To	Δ
Γ	$\mathfrak{u}$	$\mathfrak{d}$	+1
$\mathbb{H}$	$\mathbb{J}$	$\mathcal{C}$	+2
$\mathfrak{C}$	$\mathfrak{v}$	$\mathfrak{c}$	-1

The chirality promotion ($\mathbb{J} \rightarrow \mathcal{C}$) is the largest single leap, a two-step jump in Markov order. Entanglement is not a stronger correlation — it is a different memory structure. The grammar identifies this as a discrete structural promotion, not a continuous dial. I emphasize this because it is the finding that most resists assimilation into conventional QM pedagogy: entanglement changes the chirality primitive, and the chirality primitive is Markov order. Memory. Entanglement is a memory structure.

0.8 The Structural Arc: 1900–1999

The grammar reveals a progression across five phases, and the progression is not the one standard QM histories tell. The standard history is chronological: Planck, then Einstein, then Bohr, then Schrödinger and Heisenberg, then Bell. The grammar's history is structural — it groups experiments by what primitives they activate, not by what year they were performed.

0.8.1 Phase 1: The Quantum Threshold (1900–1914)

Blackbody, Photoelectric, Franck-Hertz

All three share $\uparrow$ (functorial), $\mathcal{C}$ (no symmetry), $\mathfrak{v}$ (local scope). They differ only in topology and kinetics. The grammar's signature of threshold discovery: energy quantization demonstrated through local, memoryless, functorial mappings. Nature says "energy comes in packets" and the experiments say "we confirm this by mapping photon energy to electron energy, one-to-one."

0.8.2 Phase 2: Spin and Quantization (1922)

Stern-Gerlach

The first experiment to activate $\prec$ (partial symmetry — the $\mathbb{Z}_2$ of spin-up/spin-down) and $\wp$ (disjunctive interaction grammar — the electron goes one path OR the other). This is the grammar's earliest registration of quantum discreteness as symmetry structure.

0.8.3 Phase 3: Matter Waves (1923–1927)

Compton, Davisson-Germer

Structural consolidation. These experiments share the Compton-photoelectric archetype ($\swarrow$, $\uparrow$, $\mathcal{L}$, $\mathcal{O}$, $\nearrow$, $\mathcal{I}$) with minor variations — Davisson-Germer swaps $\mathcal{I}$ for $\wp$ (conjunctive to disjunctive grammar) because the electron diffraction is an either/or path choice. Compton is identical to photoelectric. The structural consolidation phase adds no new primitives; it confirms the existing pattern across new media.

0.8.4 Phase 4: The Structural Upgrade (1927, 1959)

Double-Slit, Aharonov-Bohm

The double-slit activates three new primitives simultaneously: ν (bidirectional relational mode — the which-path and interference pattern are coupled), $\mathfrak{v}$ (quantum superposition symmetry), and $\mathfrak{l}$ (mesoscale scope — the two slits are neither local nor universal). This is the grammar's largest single structural leap within $\mathcal{J}$: three primitive promotions in one experiment.

The Aharonov-Bohm effect follows 32 years later with the grammar's most economical structural upgrade.

Aharonov-Bohm

One primitive changes: $\mathfrak{z} \rightarrow \mathfrak{s}$. That is the whole revolution. One value. The grammar's most economical upgrade. I have already discussed the difficulty of this classification. What I add here is that the structural economy of the AB effect — one primitive, one leap — is what makes it so conceptually difficult. The entire apparatus of local fields becomes insufficient, and the replacement is a single topological invariant. The grammar registers this as a one-value change because it was built to register exactly this kind of change. QM took thirty years to absorb it.

0.8.5 Phase 5: Entanglement and Nonlocality (1981–1999)

Bell Test, Quantum Eraser

The final phase introduces two new primitives simultaneously: $\mathcal{L}$ (two-step Markov memory — bipartite entanglement correlations) and either $\mathfrak{d}$ (maximal scope, Bell) or $\mathcal{I}$ (sequential grammar, Eraser). The quantum eraser activates four distinctive primitives: $\mathcal{L}$, $\mathcal{I}$, ν , and $\mathfrak{v}$. It is the structurally richest $\mathcal{J}$ experiment — the closest any QM experiment comes to the $\mathfrak{d}$ boundary while remaining firmly within $\mathcal{J}$. But it does not cross. The eraser erases path information; it does not model itself erasing. There is a structural difference between *using* delayed choice and *representing* the delayed choice within the system's own dynamics. The grammar draws this boundary at $\odot$. The quantum eraser stands at the boundary looking across.

0.9 Structural Findings

The grammar operates at $\downarrow$. The experiments it classifies operate at $\downarrow$. This tier gap means the grammar can articulate truths about QM experiments that QM cannot state about itself.

0.9.1 Structural Identity Across Parametric Difference

The Compton-photoelectric identity ($d = 0$) is invisible within QM. Different cross-sections, different energy regimes, different kinematic formulas — QM sees different phenomena. The grammar sees the same structural archetype. This is the grammar's primary mode of insight: dissolving parametric difference into structural identity.

There is a temptation to frame this as a resolution question: *is the identity real, or is it an artifact of the grammar's coarse-graining?* The temptation is wrong. The crystal of types has 17,280,000 addresses. Every system in the catalog occupies exactly one. The projections are infinite — any given address can be realized in arbitrarily many parametric media. Co-typing is identity. Two systems that share a Frobenius address are the same structural type, instantiated in different media. There is no "resolution artifact" — the grammar is not a coarse-graining of a finer continuous space; it is a structural classification whose primitives are orthogonal and discrete by construction. The parametric differences that dominate QM's self-description — cross-sections, energy regimes, kinematic formulas — are not structural primitives. They are coordinates in the medium. The grammar distinguishes medium from type. That is what it was built to do.

The photoelectric effect and Compton scattering are the same structural type. The address is the address. The distance is zero. That is the fact.

0.9.2 The ε Primitive Is Under-Utilized

Only one of ten experiments activates topological winding. The grammar predicts that $\downarrow$ systems *can* carry ε — the AB effect is not exotic but exemplary. There should be more topological quantum experiments possible within $\downarrow$. This is a structural prediction derivable from the primitive assignments.

0.9.3 The $\downarrow$ Ceiling

No QM experiment can reach $\downarrow$ without promoting $\downarrow \rightarrow \odot$. This is the structural ceiling. No amount of refinement — better detectors, lower noise, larger Hilbert spaces — crosses the boundary. The kind of system must change.

The structural requirement for $\downarrow$ is $\odot$. All ten QM experiments are at $\downarrow$, so they remain at $\downarrow$. The promotion path from $\downarrow \rightarrow \odot$ is encoded. That is the structural fact.

0.10 What the Grammar Sees

I opened with an expectation and a surprise. The expectation: all QM experiments would sit at $\downarrow$. That held. The surprise was what happens inside that floor — the Compton-photoelectric identity, the AB classification struggle, the quantum eraser at the boundary.

The grammar provides a structural microscope. Where conventional physics sees ten distinct experiments — different apparatus, different particles, different equations, different decades — the grammar sees a 12-dimensional lattice. Each experiment occupies a precise

Frobenius address. The distances between them are computed, not judged. The promotions between them are enumerated, not guessed.

The grammar's key findings:

Finding	Structural Statement
Universal floor	All QM experiments: $\downarrow, /, \downarrow, \downarrow$
Structural identity	Photoelectric $\equiv$ Compton ($d = 0$) — same archetype
Classification resistance	AB effect forced re-examination of $\downarrow$ vs $\downarrow$
Topological uniqueness	Only AB carries ε
Entanglement signature	Bell Test and Quantum Eraser carry $\mathcal{L}$
Maximal separation	Quantum Eraser $\leftrightarrow$ AB: $d = 5.86$
Consciousness ceiling	All QM experiments: $C = 0$ (Gate 1 closed)
Tier boundary	$/ \rightarrow \odot$ required to reach $\mathfrak{d}$
Structural frontier	Quantum Eraser is the richest $\downarrow$ experiment
Gate 1	Closed across all ten experiments ($C = 0$)

QM occupies $\downarrow$. The grammar, at $\downarrow$, describes QM, describes itself describing QM, and describes the structural gap between the two.

The structural requirement for $\mathfrak{d}$ is $\odot$. All ten QM experiments are at $/$ — Gate 1 is closed, consciousness score zero. The promotion from $/ \rightarrow \odot$ is encoded as a structural primitive change. That is the result.

All structural quantities were verified through tool calls to the Imscribing Grammar catalog. Tuple assignments followed the deterministic imscribing procedure [1]→[12]. The grammar's own type remains $\langle \downarrow; \mathfrak{d}; \downarrow; \mathfrak{d}; \mathfrak{d}; \downarrow; \downarrow; \downarrow; \odot; \mathcal{L}; \mathfrak{d}; \varepsilon \rangle$ — the only $\downarrow$ type capable of observing the $\downarrow$ tier from above.